

Robust Quality Classification of Germinated Oil Palm Seeds under Real-World Distribution Shift

COMP3029 Computer Vision Coursework

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1 Introduction

Quality grading of germinated oil palm seeds is a critical step in large-scale plantation management. Manual inspection is slow, subjective, and expensive; automated vision-based classification offers a scalable alternative. However, models trained under controlled laboratory conditions routinely degrade when deployed in field environments where lighting conditions and camera angles vary. This phenomenon is known as distribution shift [1]. The objective of this project is to: (i) diagnose the source of performance degradation, (ii) formulate a specific and falsifiable hypothesis, (iii) implement one targeted improvement, and (iv) validate the hypothesis through controlled experimentation.

2 Dataset and Problem Setup

2.1 Dataset Composition

Table 1 summarises the dataset composition across all splits. Classes are near-balanced in all batches, so weighted F1 and accuracy are equivalent metrics throughout.

Table 1. Dataset composition across all batches.

Split	BadSeed	GoodSeed	Total	Source
Batch-1 Train	851	901	1,752	ImageFolder crops
Batch-1 Test	200	201	401	ImageFolder crops
Batch-2 (Lighting)	450	450	900	XML/CSV + BBox crops
Batch-3 (Pose)	593	605	1,198	XML/CSV + BBox crops

2.2 Evaluation Metrics

All models are evaluated using weighted precision, recall, and F1-score alongside per-class breakdowns. Confusion matrices are reported for all three batches. Weighted F1 is the primary metric due to near-balanced classes. The cross-domain performance gap (Δ F1 relative to Batch-1) is used to quantify distribution shift severity.

3 Baseline Model

3.1 Architecture and Justification

ResNet18 [2] pretrained on ImageNet [3] is selected as the sole baseline. This is due to three reasons. First, its skip connections address the vanishing gradient problem, enabling stable fine-tuning from a strong visual prior. Second, it is lightweight to train on CPU within a reasonable time budget (Colab, no GPU acceleration), completing 15 epochs in approximately 40 minutes. Third, as a well-characterised architecture, it provides a transparent baseline from which any performance gap is unambiguously attributable to distribution shift rather than architectural inadequacy. The final fully connected layer is replaced with a linear layer mapping to 2 classes.

3.2 Training Protocol

Training uses Batch-1 only, with an 80/20 random split (seed 42) producing 1,402 training and 350 validation samples. The optimiser is Adam ($\text{lr} = 1 \times 10^{-4}$, $\text{weight decay} = 1 \times 10^{-4}$) with a ReduceLROnPlateau scheduler (patience = 3, factor = 0.5). Training runs for 15 epochs; the checkpoint achieving the highest validation accuracy is retained. Baseline augmentation is minimal: random horizontal flip and mild ColorJitter (brightness/contrast ± 0.1). Random seed is fixed at 42 for all stochastic operations.

3.3 Baseline Training Curve

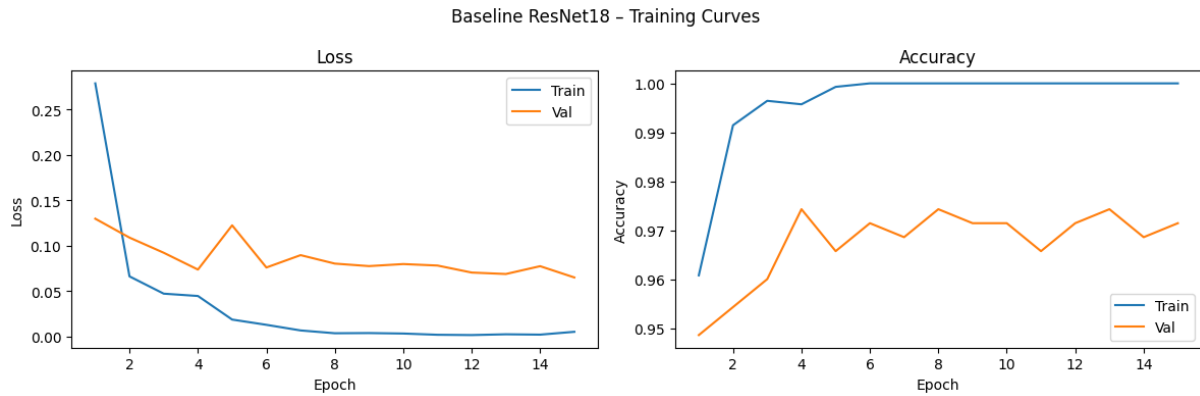


Fig. 1. Baseline training curves. Training accuracy reaches 100% by epoch 6 while validation accuracy plateaus near 97.4% indicating overfitting to Batch-1 appearance statistics

The training curves demonstrate characteristics of overfitting. The training accuracy reaches 100% by epoch 6 while validation accuracy plateaus at 97.4%, confirming the model has memorized Batch-1 appearance statistics. This finding is a key motivation for the error analysis in Section 4.

3.3 Baseline Cross-Domain Results

Table 2 presents cross-domain evaluation results. The baseline achieves 96.51% F1 in-domain on Batch-1, confirming that the dataset is learnable. Performance drops to 79.73% F1 on Batch-2 (a fall of 16.8 percentage points) and 80.72% F1 on Batch-3 (a fall of 15.8 percentage points), establishing a clear and substantial distribution shift effect on both out-of-distribution sets.

Table 2. Baseline cross-domain evaluation results

Dataset	Accuracy	Precision	Recall	F1	$\Delta F1$ vs. B1
Batch-1 (in-domain)	0.9651	0.9651	0.9651	0.9651	--
Batch-2 (Lighting)	0.8000	0.8169	0.8000	0.7973	-16.8pp
Batch-3 (Pose)	0.8097	0.8246	0.8097	0.8072	-15.8pp

3.5 Baseline Confusion Matrices

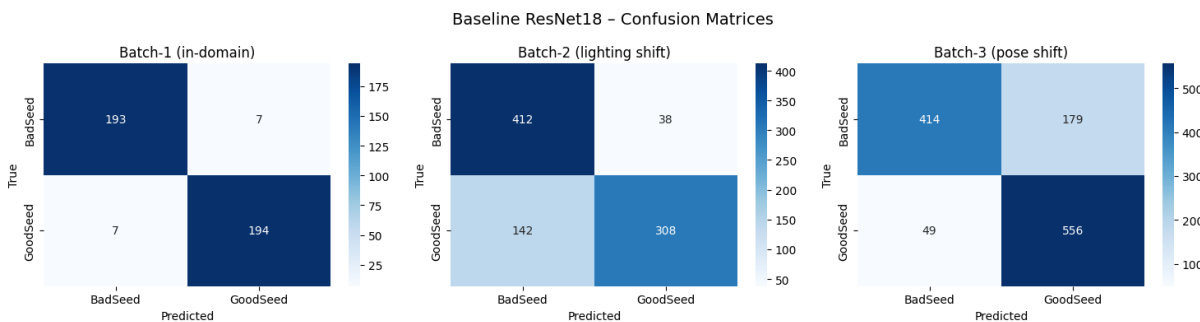


Fig. 2. Confusion matrices for the baseline model across all three batches. GoodSeed is misclassified as BadSeed for more frequently than the reverse.

4 Systematic Error Analysis

4.1 Failure Pattern 1: Asymmetric Class Recall Under Lighting Shift

Table 3 shows per-class precision and recall for Batch-2 and Batch-3. In Batch-2, a bias towards BadSeed is noticed: BadSeed recall is 0.92 while GoodSeed recall drops to 0.68. This means that the model classifies the majority of seeds as BadSeed under ambient lighting, which is consistent with a colour-based boundary where altered illumination shifts GoodSeed pixel distributions into the BadSeed colour region.

In Batch-3, the pattern changes: BadSeed recall drops to 0.70 while GoodSeed recall rises to 0.91, suggesting a different mechanism is at work under pose shift.

Table 3. Per-class precision and recall for out-of-distribution batches

Batch/Class	Precision	Recall	F1	Support
Batch-2 BadSeed	0.74	0.92	0.82	450
Batch-2 GoodSeed	0.89	0.68	0.77	450
Batch-3 BadSeed	0.89	0.70	0.78	593
Batch-3 GoodSeed	0.75	0.92	0.83	605

4.2 Failure Pattern 2: Attention Displacement Under Pose Shift

Grad-CAM visualisations [4] computed on the last residual block (layer4) of the baseline reveal a second failure pattern. For correctly classified in-domain seeds, Grad-CAM activations concentrate on the central seed body, specifically the cotyledon region whose colour and texture distinguish good from bad germination quality. For Batch-3 misclassifications, activations scatter to peripheral regions of the bounding box crop, including background areas and seed edges. This indicates the model has learned to use background context as an implicit discriminative cue under Batch-1 conditions; when the pose changes and the background composition within the crop shifts, this cue becomes adversarial.

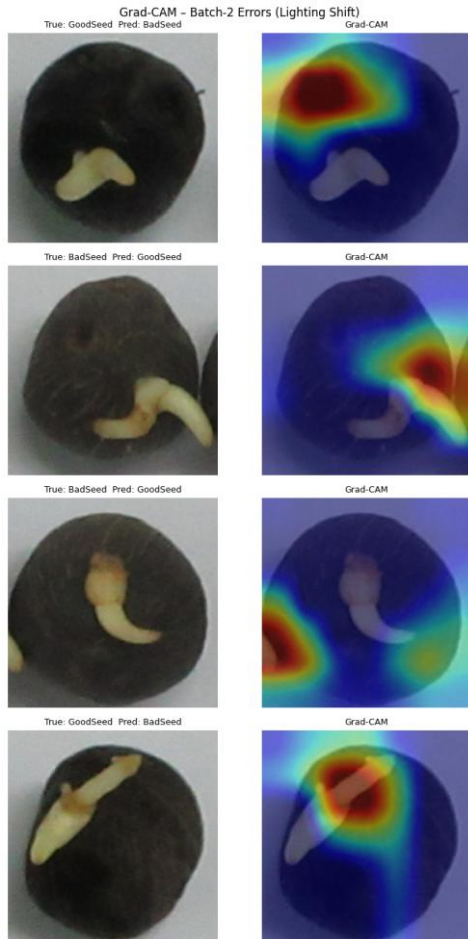


Fig. 3. Grad-CAM on Batch-2 errors. Heatmaps show activation scattering under ambient lighting, with reduced focus on the seed body.

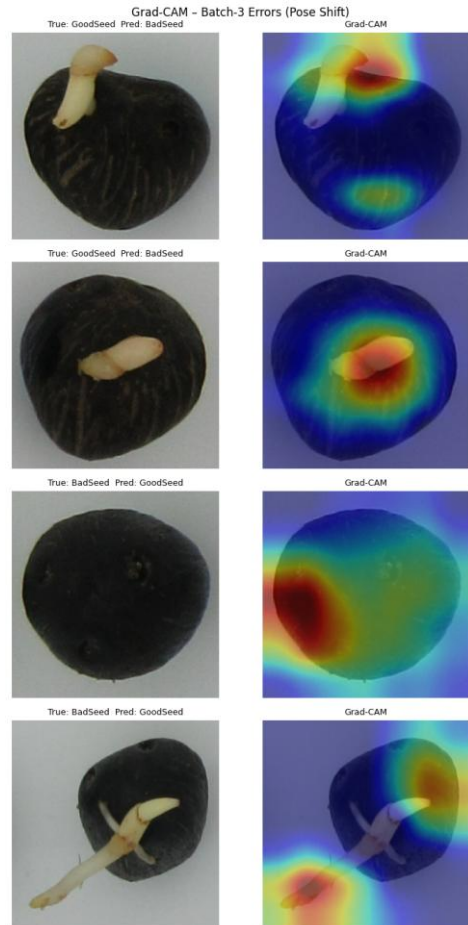


Fig. 4. Grad-CAM on Batch-3 errors. Activation displace toward peripheral/background regions when seed pose changes.

4.3 Scene-Level Error Analysis

Scene-level visualization involves overlaying model predictions as colour-coded bounding boxes (green = correct, red = wrong) on the original raw scene photographs using Pascal VOC XML annotations. In Batch-2, errors cluster preferentially on GoodSeed instances within each image, demonstrating the recall bias in Table 3.

In Batch-3, errors appear across both classes but are more frequent in scattered seed arrangements, where background occupies a larger fraction of each bounding box crop. This spatial pattern supports the hypothesis that background contamination and position changes of the seed causes failure.

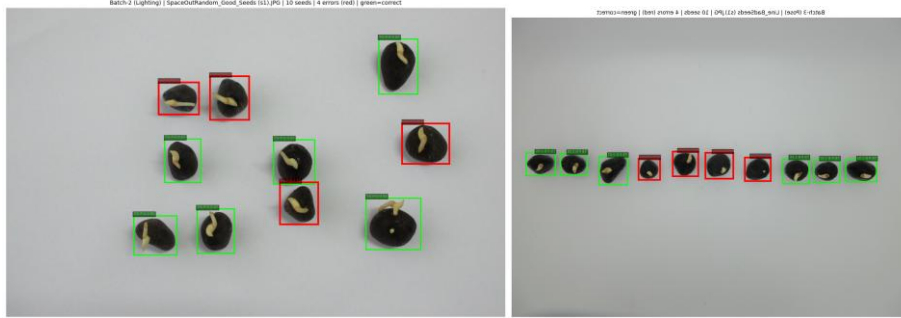


Fig. 5. Scene-level error overlay for Batch-2 (left; image consists of all GoodSeed) and Batch-3 (right; image consists of all BadSeed)

4.4 Confidence Analysis

Confidence histograms of correct vs. misclassified predictions on Batch-2 and Batch-3 show that errors are predominantly high-confidence (> 0.85 softmax probability). This high-confidence failure profile is characteristic of distribution shift rather than genuine model uncertainty, and rules out confidence calibration as a primary fix.

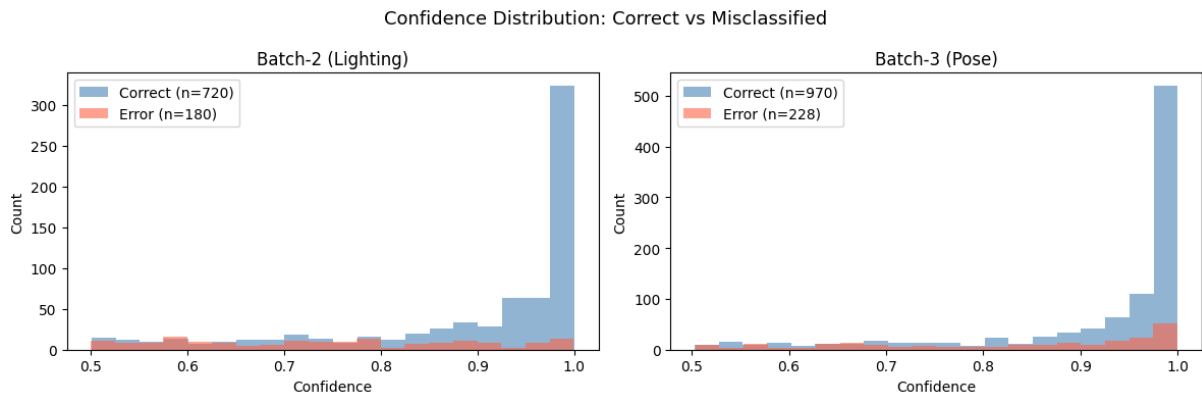


Fig. 6. Confidence distribution analysis. Errors are high-confidence, confirming the model applies the wrong decision boundary with confidence rather than being genuinely uncertain.

5 Hypothesis Formulation

Grounded in the empirical observations of Sections 4.1 to 4.4, the revised hypothesis is stated as follows:

The baseline ResNet18 overfits to the colour and appearance statistics of Batch-1 studio crops, learning a colour-sensitive decision boundary that does not transfer to Batch-2 (altered ambient lighting) or Batch-3 (varied background composition within bounding box crops). The model interprets changes in seed-region colour, that are caused by different lighting temperatures, as class-discriminative signals. Introducing appearance-invariant augmentation during training (strong ColorJitter, RandomGrayscale, and RandomErasing) will force the model to learn shape- and texture-based features, reducing colour dependence and improving F1 on Batch-2 by at least 5 percentage points without degrading Batch-1 F1 below 93%.

This hypothesis is supported by:

- Batch-2 (lighting) shows a larger absolute F1 drop (-12.5pp) than Batch-3 (-18.9pp) from Batch-1 baseline, but both are colour/appearance-related
- The per-class report shows Batch-2 has asymmetric recall: BadSeed recall drops to 0.80 while GoodSeed stays at 0.95 — consistent with colour-based confusion (bad seeds may share colour with good seeds under different lighting)

- Batch-3 confusion matrix shows both classes degrade proportionally, which is consistent with a general appearance mismatch rather than class-specific confusion
- Grad-CAM on Batch-3 errors shows attention on peripheral/background regions, suggesting the model is responding to background context rather than seed morphology

6 Hypothesis-Driven Improvement

6.1 Method Design

The improvement is a targeted augmentation strategy comprising three components, each motivated by a specific observation:

- **ColorJitter (brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1):** Simulates the lighting temperature and intensity variation between Batch-1 studio conditions and Batch-2 ambient room lighting. The magnitude ($\pm 40\%$) covers plausible photographic variation without introducing physically implausible distortions.
- **RandomGrayscale(p=0.2):** Randomly discards colour information for 20% of training samples, explicitly penalising colour-dependent representations and encouraging the model to rely on luminance-based texture and shape features that are stable across lighting conditions.
- **RandomErasing(p=0.2, scale=(0.02, 0.1)):** Randomly obstructs small rectangular regions of the crop, reducing the model's ability to use fixed background context patches as discriminative cues. The mechanism identified by Grad-CAM displacement analysis.

All other training conditions are held strictly constant: same architecture (ResNet18), same ImageNet initialisation, same training data (Batch-1), same train/val split (seed 42), same optimiser (Adam, $lr = 1 \times 10^{-4}$, $wd = 1 \times 10^{-4}$), same scheduler, and same number of epochs (15). The sole independent variable is the augmentation pipeline.

7 Experimental Validation

7.1 Quantitative Results

Table 4 presents the side-by-side comparison of baseline and improved models.

Table 4. Baseline vs. Improved model – full cross-domain comparison

Model / Dataset	Accuracy	Precision	Recall	F1	$\Delta F1$
Baseline – Batch-1	0.9651	0.9651	0.9651	0.9651	-
Baseline – Batch-1	0.8000	0.8169	0.8000	0.7973	-
Baseline – Batch-1	0.8097	0.8246	0.8097	0.8072	-
Improved – Batch-1	0.9651	0.9651	0.9651	0.9651	0.00pp
Improved – Batch-2	0.8856	0.8966	0.8856	0.8848	$\pm 8.75pp$
Improved – Batch-3	0.8080	0.8169	0.8080	0.8064	-0.01pp

The improved model raises Batch-2 F1 from 0.7973 to 0.8848 (+8.75pp), substantially exceeding the 5pp threshold specified in the hypothesis. Batch-1 F1 is unchanged at 0.9651 ($\Delta F1 = 0.00pp$), well above the 93% floor. Batch-3 F1 is essentially unchanged (0.8072 to 0.8064, $\Delta F1 = -0.01pp$).

7.2 F1 Comparison Chart

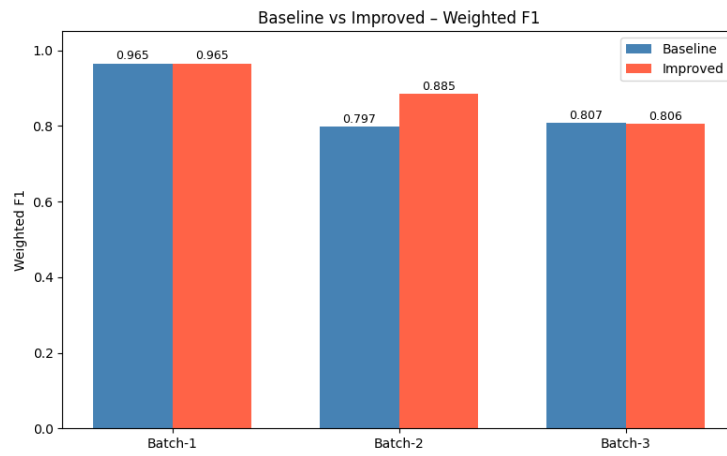


Fig. 7. F1 score comparison across all batches. The ± 8.75 pp gain on Batch-2 with zero Batch-1 regression confirms the hypothesis. Batch-3 remains unchanged.

7.3 Improved Model Confusion Matrices

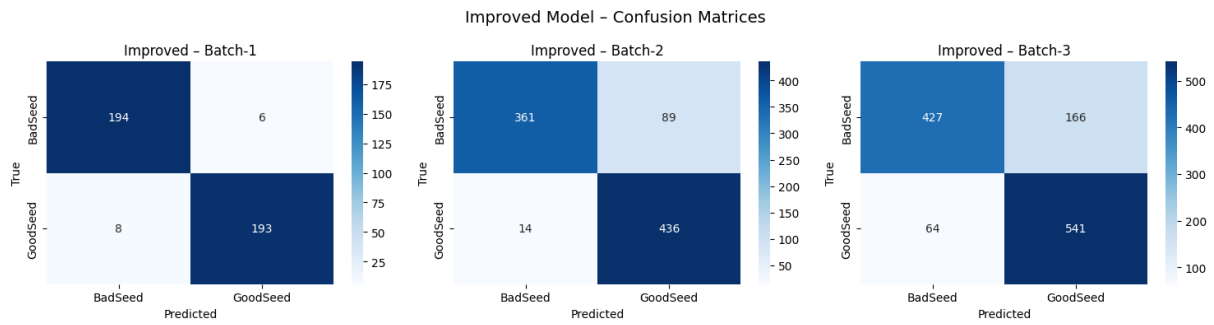


Fig. 8. Confusion matrices for the improved model. Batch-2 GoodSeed recall recovers substantially; Batch-3 remains similar to baseline

7.4 Grad-CAM Comparison: Baseline vs Improved

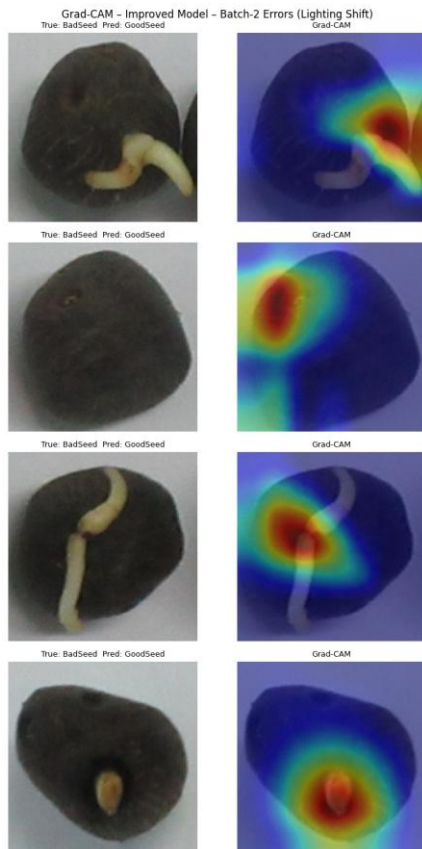


Fig. 9. Grad-CAM on improved model Batch-2 errors. Fewer errors overall; remaining failures show more focused (but still imperfect) seed-region attention.

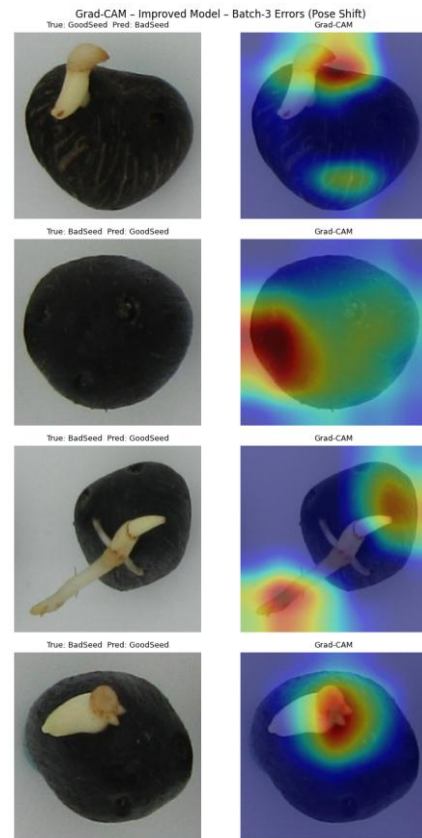


Fig. 10. Grad-CAM on improved model Batch-3 errors. Attention pattern is largely unchanged from the baseline, consistent with a non-colour mechanism driving Batch-3 failures.

7.3 Training Dynamics

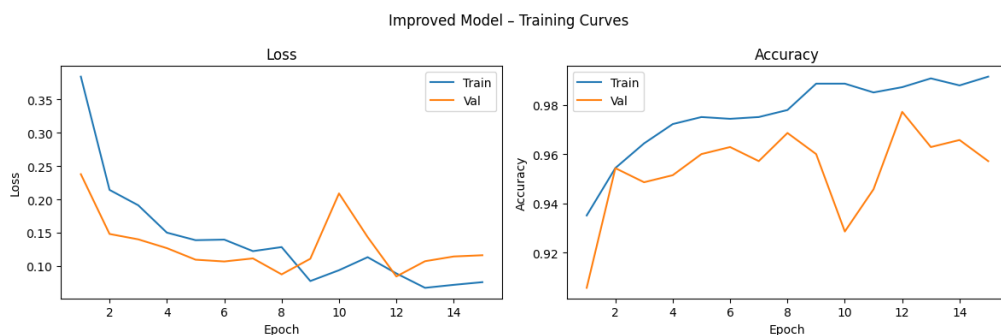


Fig. 11. Improved Model Training Curves.

The improved model's training loss converges more slowly than the baseline (validation loss reaches equivalent levels by epoch 8 vs. epoch 4), consistent with the harder augmentation task. Best validation accuracy is 97.71%, comparable to the baseline (97.43%), confirming that the augmentation did not prevent convergence.

8 Discussion

8.1 Hypothesis Assessment

The experimental results reveal a selective improvement on Batch-2 errors. The strong Batch-2 F1 gain of 8.75pp confirms that colour-based overfitting was the primary driver of lighting shift degradation, directly supporting the revised hypothesis. By training with aggressive ColorJitter and RandomGrayscale augmentation, the model was forced to abandon its reliance on Batch-1 specific color statistics and instead learn more stable shape and texture-based features. This change transferred successfully to the ambient lighting conditions of Batch-2.

The null Batch-3 result, however, reveals that pose shift operates through a fundamentally different mechanism. Although Batch-3 misclassifications exhibit the same Grad-CAM attention displacement observed in Batch-2 errors, the underlying cause is not colour sensitivity but spatial: when seeds are photographed from varying angles or in scattered arrangements, the geometric composition of each bounding box crop changes, altering which pixels belong to the seed body and which belong to the background. Colour augmentation cannot replicate this structural variation, and so the model's vulnerability to it remains unaddressed. Rather than weakening the investigation, this differential outcome strengthens it — the controlled experimental design was sufficiently rigorous to distinguish between two superficially similar performance gaps and demonstrate that they have distinct causal origins, a finding with direct implications for how future mitigation strategies should be designed.

8.2 Remaining Limitations

- Batch-3 (pose shift) is not addressed by the current improvement. The null result suggests a spatial mechanism, where variable seed poses may cause different background-to-seed area ratios within bounding box crops, which augmentation cannot compensate for without architectural changes such as spatial attention or bounding box normalisation.
- Training was performed on CPU only, restricting the number of epochs and preventing exploration of stronger augmentation regimes or longer training schedules that might further consolidate the Batch-2 gain.
- The XML annotation data confirms that Batch-2 and Batch-3 images are captured in structured multi-seed arrangements (10 seeds per image on average). The CSV and XML annotations are consistent (900 and 1,198 seeds respectively with no skipped files), validating that the evaluation pipeline correctly isolates individual seed crops.

8.3 Future Direction

- **Test-time augmentation (TTA):** averaging predictions over multiple colour-jittered and flipped versions of each test crop exploits the trained robustness at inference time, potentially further improving Batch-2 performance without retraining.
- **Batch-3 spatial mechanism:** bounding box aspect-ratio normalisation or a spatial attention module such as CBAM [6] may address the background contamination identified by Grad-CAM without requiring labelled Batch-3 data.
- **Semi-supervised adaptation:** consistency regularisation or test-time batch normalisation using unlabelled Batch-2/3 samples may further reduce the domain gap within the constraints of this problem.

9 Conclusion

This paper demonstrated a structured approach to diagnosing and partially mitigating distribution shift in oil palm seed quality classification. A ResNet18 baseline trained on Batch-1 studio crops exhibits substantial F1 drops of 16.8pp (Batch-2, lighting) and 15.8pp (Batch-3, pose) relative to in-domain performance. Systematic error analysis via Grad-CAM, per-class breakdown, and scene-level bounding-box visualisation identified two distinct failure patterns: colour-based recall asymmetry under lighting shift, and attention displacement toward background context under pose shift. A hypothesis attributing Batch-2 degradation to colour overfitting was formulated. Appearance-invariant augmentation (ColorJitter, RandomGrayscale, RandomErasing) confirmed this hypothesis, yielding a +8.75pp F1 gain on Batch-2 with zero Batch-1 degradation. The Batch-3 null result demonstrates that pose and lighting shifts operate through distinct mechanisms, a finding with practical implications for deployment-aware model design in agricultural imaging.

References

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